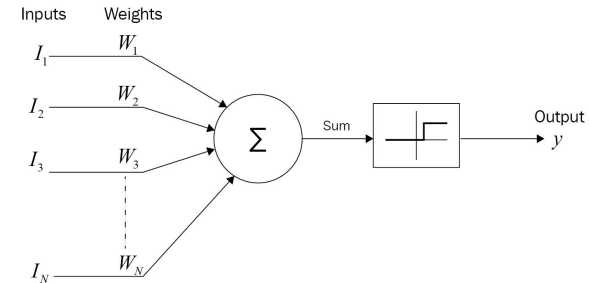




Perceptron

Weighted Networks - Perceptron

- McCulloch–Pitts (MP) - build networks capable of computing any logical function.
- Computing units are too similar to conventional logic gates
- Network needs to be pre-specified
- No free parameters [adaptable to suit different problems]
- Learning can only be implemented by modifying the connection pattern of the network and the thresholds of the units, but this is necessarily more complex than just adjusting numerical parameters.



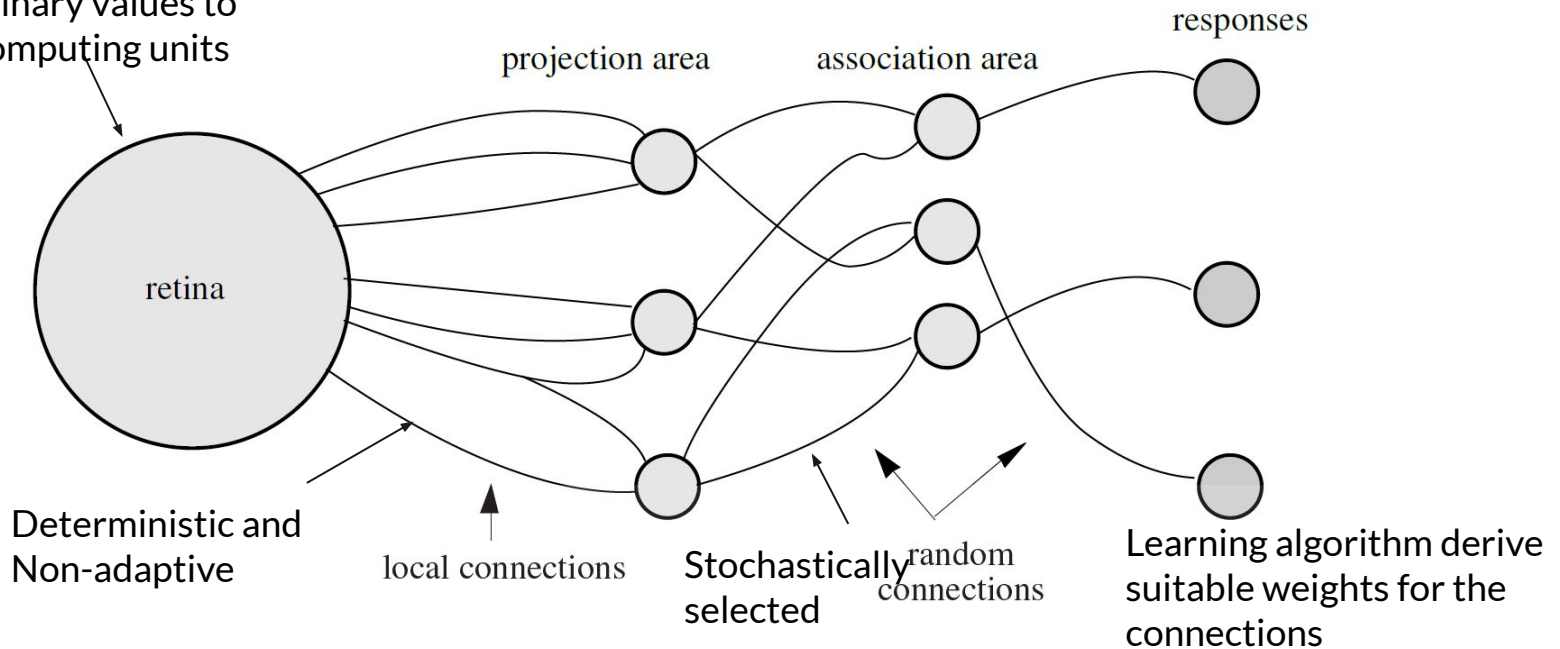


Weighted Networks - Perceptron

- In 1958 Frank Rosenblatt proposed the perceptron a more general computational model than McCulloch–Pitts units.
- Innovation was the introduction of numerical weights and a special interconnection pattern.
- In Rosenblatt model the computing units are threshold elements and the connectivity is determined stochastically
- Learning takes place by adapting the weights of the network with a numerical algorithm.
- Minsky and Papert refines the Rosenblatt model to be a perceptron

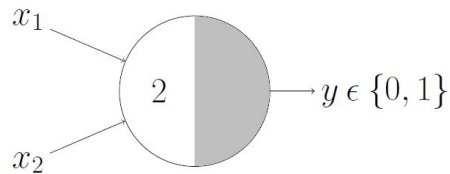
The classical perceptron [Rosenblatt]

Transmits binary values to
a layer of computing units



MP Vs Rosenblatt model

1. Presence of weights in the network [Rosenblatt model]
2. limit on the maximum acceptable fan-in of the units [Rosenblatt model]
3. MP Neuron Model only accepts boolean input whereas Perceptron Model can process any real input.



AND function

$$x_1 + x_2 = \sum_{i=1}^2 x_i \geq 2$$



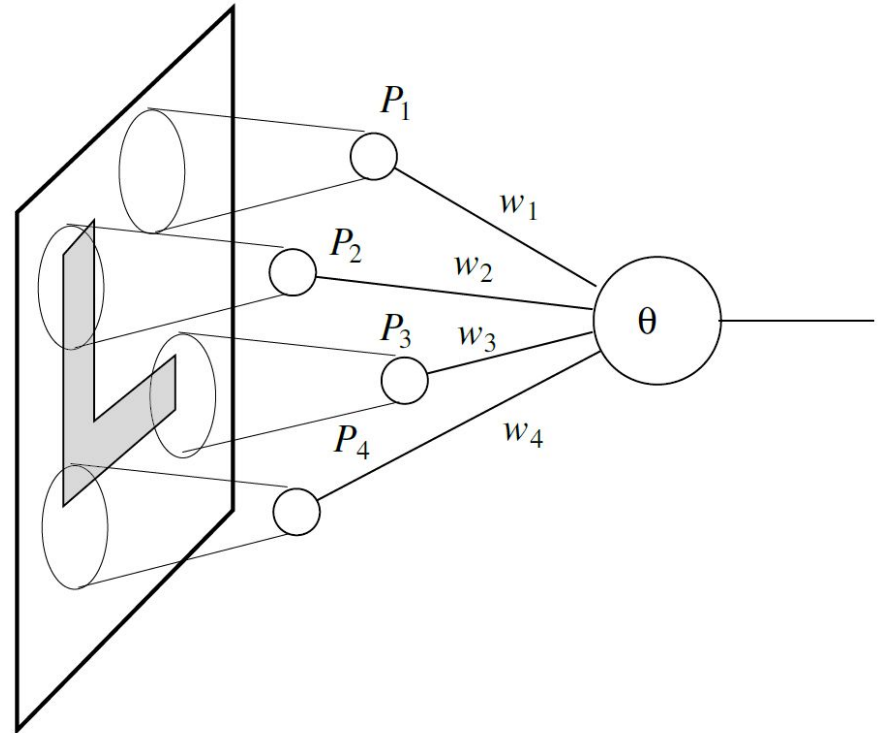
Minsky and Papert Vs Rosenblatt model

- Minsky and Papert distilled the essential features from Rosenblatt's model in order to study the computational capabilities of the perceptron under different assumptions.
- Retina of pixels with binary values on which patterns are projected.
- Some pixels from the retina are directly connected to logic elements called predicates which can compute a single bit according to the input.
- Interestingly, these predicates can be as computationally complex
- The predicates transmit their binary values to a weighted threshold element which is in charge of reaching the final decision in a pattern recognition problem.

Predicates Model [Minsky and Papert]

- The predicates P_1 to P_4 deliver information about the points in the projection surface that comprise their receptive fields.
- The only restriction on the computational capabilities of the predicates is that they produce a binary value and the receptive field cannot cover the whole retina.
- The threshold element collects the outputs of the predicates through weighted edges and computes the final decision.

$$\sum_{i=1}^n w_i P_i \geq \theta$$





Computational limits of the perceptron model

- Minsky and Papert used their simplified perceptron model to investigate the computational capabilities of weighted networks.
- Early experiments with Rosenblatt's model had aroused unrealistic expectations in some quarters, and there was no clear understanding of the class of pattern recognition problems which it could solve efficiently.
- To explore this matter the number of predicates in the system is fixed, and although they possess unbounded computational power, the final bottleneck is the parallel computation with a single threshold element.
- This forces each processor to cooperate by producing a partial result pertinent to the global decision.



Amdahl's law

- The system considered by Minsky and Papert at first appears to be a strong simplification of parallel decision processes, but it contains some of the most important elements differentiating between sequential and parallel processing.
- It is known that when some algorithms are parallelized, an irreducible sequential component sometimes limits the maximum achievable speedup.
- The mathematical relation between speedup and irreducible sequential portion of an algorithm is known as Amdahl's law.
- In the model considered above the central question is, are there pattern recognition problems in which we are forced to analyze sequentially the output of the predicates associated with each receptive field or not?
- Minsky and Papert showed that problems of this kind do indeed exist which cannot be solved by a single perceptron acting as the last decision unit.

Perceptron types [Connection to predicates]

- Diameter limited perceptrons: the receptive field of each predicate has a limited diameter.
- Perceptrons of limited order: each receptive field can only contain up to a certain maximum number of points.
- Stochastic perceptrons: each receptive field consists of a number of randomly chosen points



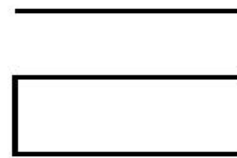
A



B



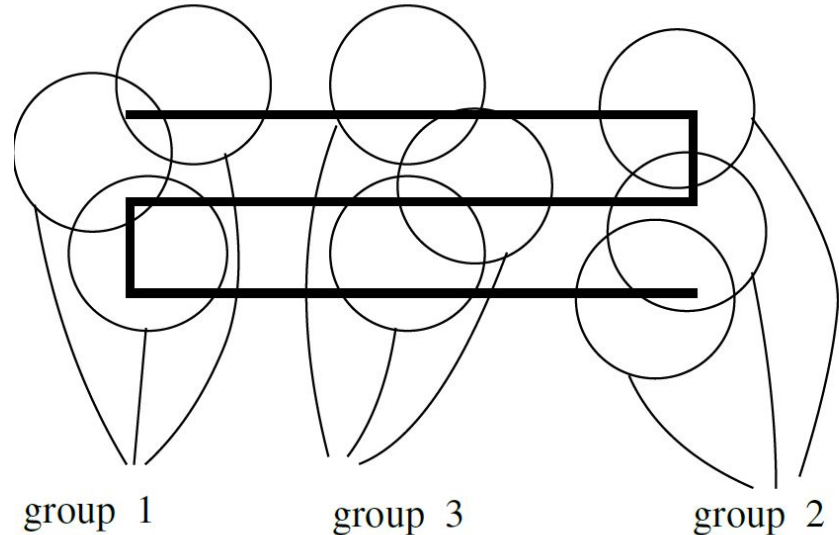
C



D

Diameter limited perceptron

- The first group consists of those predicates whose receptive fields contain points from the left side of a pattern.
- Predicates of the second group are those whose receptive fields cover the right side of a pattern.
- All other predicates belong to the third group.



Diameter limited perceptron

- If S is positive the figure is recognized as connected

$$S = \sum_{P_i \in \text{group 1}} w_{1i} P_i + \sum_{P_i \in \text{group 2}} w_{2i} P_i + \sum_{P_i \in \text{group 3}} w_{3i} P_i - \theta \geq 0.$$

- If the disconnected pattern A is analyzed, then we should have $S < 0$.

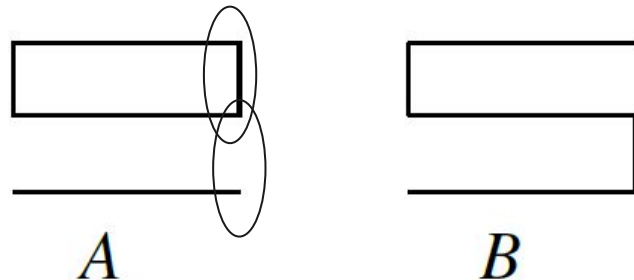


A

Diameter limited perceptron

- Pattern A can be transformed into pattern B without affecting the output of the predicates of group 3 and 1
- The predicates of group 2 adjust their outputs by $\Delta_2 S$

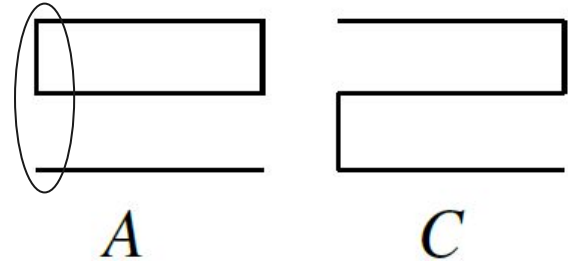
$$S + \Delta_2 S \geq 0 \Rightarrow \Delta_2 S \geq -S.$$



Diameter limited perceptron

- If pattern A is transformed into pattern C, the predicates of group 1 adjust their outputs so that the threshold element receives a net excitation

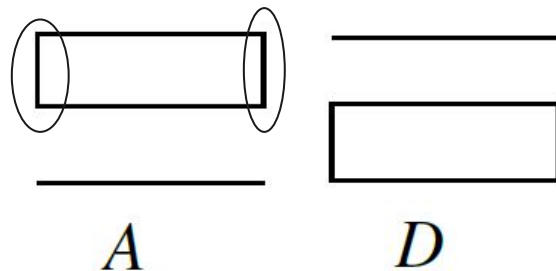
$$S + \Delta_1 S \geq 0 \Rightarrow \Delta_1 S \geq -S.$$



Diameter limited perceptron

$$\Delta S = \Delta_2 S + \Delta_1 S \geq -2S,$$

$$S + \Delta S \geq -S > 0$$





Diameter limited perceptrons

- The connectedness of a figure is a global property which cannot be decided locally.
- If no predicate has access to the whole figure, then the only alternative is to process the outputs of the predicates sequentially.



Implementation of logical functions